Cognizant Academy

Generation Cognizant (GenC) Learner Handbook (Quality Engineer with Selenium Automation – v3)



Why do we need this GenC learning Program?

Gen C learning program engages young talents with a comprehensive learning pathway, giving the millennials an opportunity to interact with Subject Matter Experts (SME), understand the corporate environment, and groom themselves.

Cognizant emphasizes on Learner Autonomy where students take charge of their own learning, with the available tools and resources. More focus is on “learning” than “teaching”. Get ready to embark your own learning adventure!

Program at a glance

Learning consisting of 2 Stages:

- Stage 1 – Core Java with SQL
- Stage 2 – Functional Testing With Selenium Automation Concepts – Inclusive of the Project

  Business Aligned Project

  Interim Project Evaluation + Technical Evaluation

  Final Project Evaluation + Final Technical Evaluation

Program Highlights

- The complete learning journey is formalized using adult learning principles, where problem solving and applying the skills gained are given more importance than conceptual learning.
- Learner Autonomy is encouraged via Flipped Classroom, where the learning platform offers world class learning resources, and students would not be constrained by tutelage of an instructor.
- Get mentored by SME, whose motivation and guidance will help you accelerate in the learning journey.
- This program is applicable to Interns as well as GEN Cs.

Cognizant Quality Engineering & Assurance (QEA) focuses on ensuring the quality and reliability of enterprise processes, applications, and systems. QEA offers a comprehensive range of services including intelligent and automated quality assurance, modernization assurance, and experience assurance. These services are designed to accelerate business and technology changes, improve customer experiences, and ensure regulatory compliance. QEA also emphasizes the importance of automation and AI to enhance testing efficiency and deliver high-quality outcomes. By leveraging these advanced technologies, QEA helps businesses achieve digital success and maintain competitive advantage in the evolving digital landscape.

How QEA Transforms Quality Assurance at Cognizant?

Cognizant's Quality Engineering & Assurance (QEA) significantly transforms quality assurance by accelerating digital transformation and ensuring robust, reliable applications and systems. This seamless transition minimizes disruptions and speeds up technology adoption.

By embedding quality at every development stage, QEA enhances customer satisfaction and loyalty through high-performance, user-friendly products. Automation and AI-driven quality assurance streamline processes, reducing manual efforts and speeding up release cycles, which boosts operational efficiency and cuts costs. Rigorous testing and validation ensure compliance with industry regulations, crucial for sectors like healthcare and finance.

QEA also supports innovation by providing a reliable foundation for launching new products and entering new markets. Additionally, automated testing frees employees to focus on strategic tasks, enhancing productivity and job satisfaction. These combined impacts help businesses achieve strategic goals, maintain a competitive edge, and deliver exceptional value to their customers.

Notable & Successful Stories on Quality Assurance by QEA Cognizant Team…

Cognizant's Quality Engineering & Assurance (QEA) team partnered with a major health plan provider to overhaul their software quality assurance processes, addressing several critical challenges. The existing methods were outdated and inefficient, struggling to keep pace with the healthcare industry's evolving needs.

Ensuring compliance with stringent healthcare regulations was a significant concern, and the manual, fragmented processes led to delays and increased operational costs. Additionally, there was a pressing need to improve software quality to enhance patient care and overall service delivery.

QEA implemented automated testing frameworks and AI-driven quality assurance tools, significantly reducing manual efforts and accelerating the testing process. Rigorous testing and validation processes were established to ensure compliance with healthcare regulations, including continuous monitoring and real-time compliance checks.

By streamlining quality assurance processes and integrating advanced technologies, QEA improved the overall efficiency of the software development lifecycle.

The focus on quality from the start ensured that end products were reliable and user-friendly, leading to better patient care and satisfaction. As a result, the health plan provider saw a significant improvement in software quality and compliance, which was crucial for meeting healthcare regulations.

The enhancements in software quality directly contributed to better patient care and service delivery, while the automation and streamlined processes led to faster release cycles and reduced operational costs.

The time-to-market for software updates was significantly reduced, allowing the client to respond more quickly to market demands. This project exemplifies how Cognizant's QEA services can transform quality assurance processes, ensuring high-quality outcomes.

Health plan rebuilds software QA with Cognizant Case Study | Cognizant

Tips for Successfully Carrying Best Practices for Implementing and Maintaining Robust IT Infrastructure with QEA.

Understand the Basics: Start with a solid understanding of software development and testing fundamentals. Familiarize yourself with key concepts such as the software development lifecycle (SDLC), different types of testing (e.g., unit, integration, system, and acceptance testing), and the importance of quality assurance.

Learn Automation Tools: Gain proficiency in popular automation tools like Selenium, JUnit, and TestNG. Automation is crucial for efficient testing, and being skilled in these tools will enhance your ability to create reliable and repeatable tests.

Adopt the Page Object Model (POM): Use the Page Object Model design pattern to create maintainable and reusable code. This approach separates test logic from page elements, making your tests more organized and easier to manage.

Focus on Continuous Integration/Continuous Deployment (CI/CD): Understand and implement CI/CD practices. Tools like Jenkins help integrate testing into the development pipeline, ensuring that code changes are tested continuously and deployed seamlessly.

Embrace Agile and DevOps Practices: Agile and DevOps methodologies promote collaboration, speed, and quality. Learn how to work in these environments, focusing on continuous improvement and adaptability.

Prioritize Test Coverage: Develop comprehensive test plans that cover all aspects of the application, including functional, performance, security, and usability testing. Ensure that your tests are thorough and cover various scenarios.

Leverage AI and Machine Learning: Explore how AI and machine learning can enhance testing processes. These technologies can help predict potential issues, optimize test cases, and improve the overall effectiveness of quality assurance.

Implement Data-Driven Testing: Use data-driven testing to run the same test with different sets of data. This approach increases test coverage and ensures that your application works correctly with various inputs.

Maximize Browser Window: Ensure that your tests maximize the browser window to capture full-page screenshots and interact with all elements. This practice helps improve test accuracy and reliability.

Validate with Assertions: Use assertions to validate the outcomes of your tests. Assertions help ensure that the application behaves as expected and quickly identify any issues.

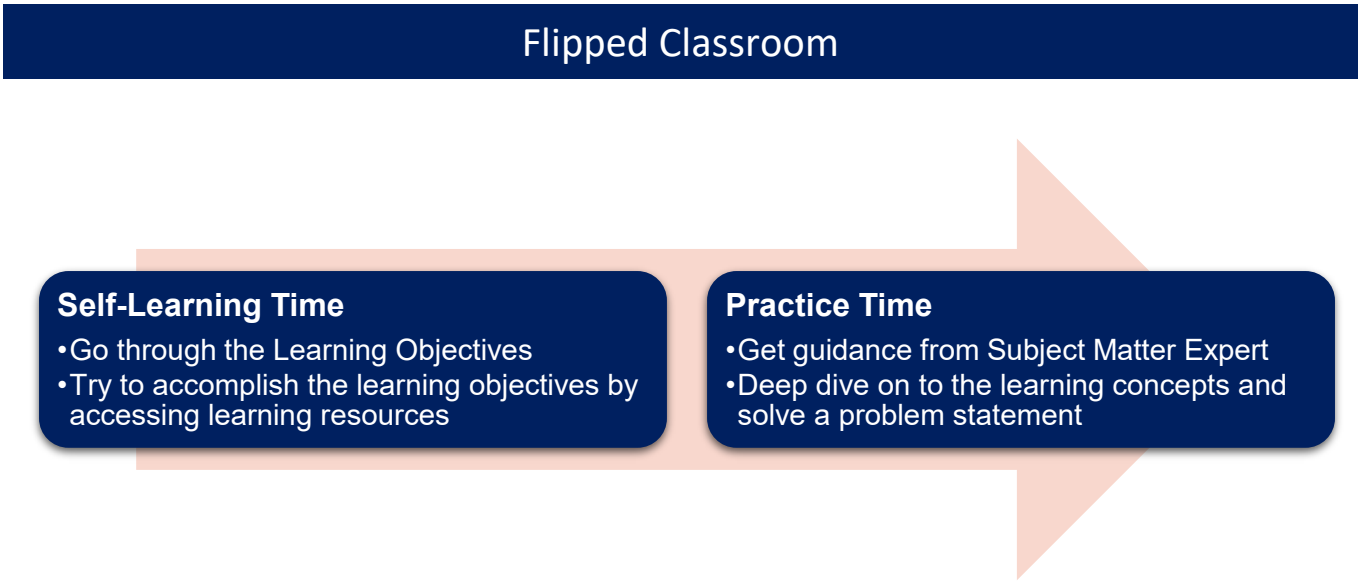
Continuous Learning and Improvement: Stay updated with the latest trends, tools, and best practices in quality assurance. Participate in training sessions, workshops, and online courses to continuously enhance your skills.

Collaborate and Communicate: Foster strong communication and collaboration with development, testing, and operations teams. Quality assurance is a shared responsibility, and effective teamwork is essential for success.

Learning Journey with Flipped Classroom

This program encourages you to be more autonomous learners during guided self-learning hours, completing the learning objectives on your own pace and style, and get ready for the hands-on practice time.

The complete learning path is set in the GEN C Learn Platform, which you can login with SSO.



Recommended Program Sequence

The learning journey contains 2 stages, followed by a Business Aligned Project.

- Stage 1 – Core Java with SQL
- Stage 2 – Functional Testing with Selenium Automation Concepts
- Business Aligned Project will provide you an experience of real time problem solving in Agile methodology.

Stage 1 – Core Java with SQL

Day 1 to 15

- Core Java
- Code Challenge (CC) Assessment
- ICT - JAVA(Integrated Capability Test) Assessment

Day 16 to 19

- SQL
- Code Challenge (CC) Assessment
- Stage 1 Mock

Day 20 & 21

- **Stage 1 Assessment**

Stage 2 – Functional Testing with Selenium Automation Concepts

Day 22 to 26

- Functional Testing
- Gen AI Learnings
- ICT - Functional Testing(Integrated Capability Test) Assessment

Day 27 to 31

- Automation Concepts, Selenium configuration, WebDriver Basics
- Code Challenge (CC) Assessment

Day 32 to 36

- Selenium Automation Techniques, Dynamic XPath
- Code Challenge (CC) Assessment
- Project Deliverables

Day 37 to 39

- Selenium Webdriver With POM and ApachePOI
- Code Challenge (CC) Assessment
- Project Deliverables

Day 40 to 45

- Automation Testing - Selenium with TestNG(With XML & JSON)
- Code Challenge (CC) & ICT Assessment
- **Interim Project Evaluation(Integrated project, Mini Project) + Technical Evaluation**

Day 46

- Automation Framework

Day 47 to 54

- Devops(GIT,Maven,Grid,Jenkins)

Day 55 to 56

- BDD Cucumber
- Project Deliverables

Day 57 to 61

- POSTMAN
- API Automation with RESTASSURED
- Project Deliverables

Day 62

- Devops Tools - Docker & Container
- Code Companion Tool- GitHubCopilot
- Project Deliverables

Day 63 to 65

- **Final Project Evaluation and Final Technical Evaluation**

Key Learning Components of the Program

Cognizant has collaborated with Percipio to provide world class learning videos for the evolving future of work. These Percipio programs are woven into a learning path, empowering you to plan and learn at your style.

The program also connects you with Subject Matter Experts (SMEs) to get the professional guidance on your queries in the learning journey.

The program doesn't ONLY concentrate on the technical skilling, but also on the shaping up of the Behavioral skills. **Behavioral learning** would be done in ILT mode, with few Self-paced learning modules too.

Evaluation Model

The program continuously evaluates if you can apply those self-learnt skills to solve a real-time business problem. Depicted below are the four key learning components, which are distributed across the learning journey for the purpose of continuous evaluation.

- Interim Evaluation (Project + Technical) through Video Interview
- Final Evaluation (Project + Technical) through Video Interview

The above evaluation components will attribute to the Performance Health Status (PHS) of a GenC. Additional Learning Components like Hands-On, Code Challenges and ICTs will help you to enhance your expertise level.

Interim & Final Evaluation Approach

Below is the Evaluation Structure for GenC Learning Journey

The interim evaluation will be held halfway through the learning journey, while the final evaluation will take place at the end of the learning journey.

During the interim evaluation, the GenC will be interviewed by a Technical Subject Matter Expert (SME) from the Business Unit (BU) to assess your knowledge through a technical discussion. Additionally, the Mini project completed by the GenC will also be evaluated by the BU SME. Please note that the Mini project is an individual activity and not a group activity.

During the final evaluation, the GenC will again be interviewed by a Technical SME from the BU to assess their knowledge through a technical discussion. In the same evaluation call, the Hackathon project completed by each group will be evaluated by the BU SME. Please note that the Hackathon project is a group activity and not an individual activity.

Preparatory Learning

For this automation track training, here are some preparatory learning materials that can be helpful for the interim and final technical evaluations. Preparatory learning material refers to resources, materials, and content that you can use to prepare. These materials are designed to provide you with the necessary knowledge, skills, and understanding to succeed in your learning objectives.

Remember, apart from these learning materials, actively participating in the training sessions, asking question and practicing hands-on exercises will greatly contribute to your overall learning and preparedness for the evaluations.

Program Completion Criteria

Stage 1 (Pre-requisites) Gating Criteria: Qualifier Assessment			
Stage 2 & Beyond (Advanced Skills) **Gating Criteria: Performance Health Status is Green**			
GenC/Intern Training	**Evaluation Components**	**Pass Criteria**	**Evaluation Done by**
Performance Health Status - PHS (only from Stage 2)	Interim Evaluation (Project + Technical)	Green,1 Attempt	BU SME
	Final Evaluation (Project + Technical)	Green,2 Attempts*	

nt and 100% Completion of Hands On in Stage 2 is mandatory for interim / final evaluation eligibility.

Key Check Point Intervals in the Learning Journey

RAG as PHS (Performance Health Status)
The program continuously evaluates if you are able to apply those self-learnt skills to solve a real time business problem. Depicted below are the two key learning components, which are distributed across the learning journey for the purpose of continuous evaluation.

Interim Evaluation:

- During the interim evaluation, the GenC will undergo a video interview on the learning platform.
- This interview will be conducted by a Tech SME from the BU.
- The purpose of this evaluation is to assess the GenC's knowledge and understanding of the skills covered in the training program up to the halfway point.
- It also encompasses an evaluation of the GenC's progress in their Project case study.
- The evaluation will involve a technical discussion as well as an assessment of the Project progression to gauge the GenC's proficiency in the skills learned thus far.
- There will be ONLY one attempt.

Final Evaluation:

- For the final evaluation, the GenC will participate in a video interview conducted by a Tech SME from the BU.
- This evaluation aims to assess the GenC's knowledge and expertise in all the skills covered throughout the entire training program.
- Similar to the interim evaluation, this assessment will involve a technical discussion via a video interview on the learning platform, along with a project evaluation to assess the GenC's capabilities and their Project case study progress.
- It serves as a comprehensive evaluation of the GenC's skills and capabilities acquired during the training.

- In Final Evaluation , Other 1 attempt is given when student PHS is AMBER/RED. (Attempt 1 is not applicable if the student is **Red** in Interim and Final Evaluation - Attempt 1)

The above evaluation components will attribute to the Performance Health Status (PHS) of a GenC. Additional Learning Components like Hands-On, Quizzes, CCs, and ICTs will help you to enhance your expertise level.

Mandatory Hands-On Exercise Completion :

- Completion of 100% of the hands-on exercises is mandatory to qualify for the stage 1 qualifier, interim, and final evaluations.

Table - Check Point Intervals

Check points	Interpreting Status
Interim Evaluation	- Green - On Track for Graduation - Red /Amber - There will not be any re-attempts given
Final Evaluation	- Green - On Track for Graduation - Red /Amber – Only 2 attempts are given - (Attempt 2 is not applicable if the student is **Red** in Interim and Final Evaluation- Attempt 1) **Note:** If student fails after the applicable re-attempts, they will be considered as “Not-Graduated”.

Icebreaker



Icebreaker session will be conducted for a duration of initial **5 days.** During the session, various topics related to Corporate Induction, Talent Management, Cognizant Agenda on Core Values, Leader Talks, Alumni, BU Mentor connects will be covered. Followed by icebreaker, technical training will kick start.

Following sessions will be covered during the **5 days** of icebreaker

- Corporate Induction
- Talent Manager Connect

- Cognizant Agenda Sessions on Core Values
- Leader Talks (Academy) and many more…

A recommended day-wise schedule is provided below for the learning, with the learning content for the day, the practice hands-on and extended hands-on to be done for the day or

any other activities are listed. Few days might be interleaved to accommodate the extension due to Behavioral Training.

Mock Interview Session: This session is conducted to test candidate understanding and enhance candidates' best preparation for Interim & Final Evaluations.

GenAI-Program Overview

Introduction

AI Accelerate is a comprehensive program designed to empower learners with the knowledge and skills needed to harness the transformative potential of Artificial Intelligence (AI). This handbook serves as a guide, providing essential information and resources to help learners navigate through the program successfully. From understanding the fundamentals of AI to apply the advanced concepts in real-world scenarios. AI Accelerate is your gateway to unlock the power of AI and shaping the future of technology.

Program overview.

AI Accelerate offers a learning opportunity, allowing GenCs to engage learning through the self-paced learning, Expert connect, and Knowledge assessment to measure the skills.

Focus areas

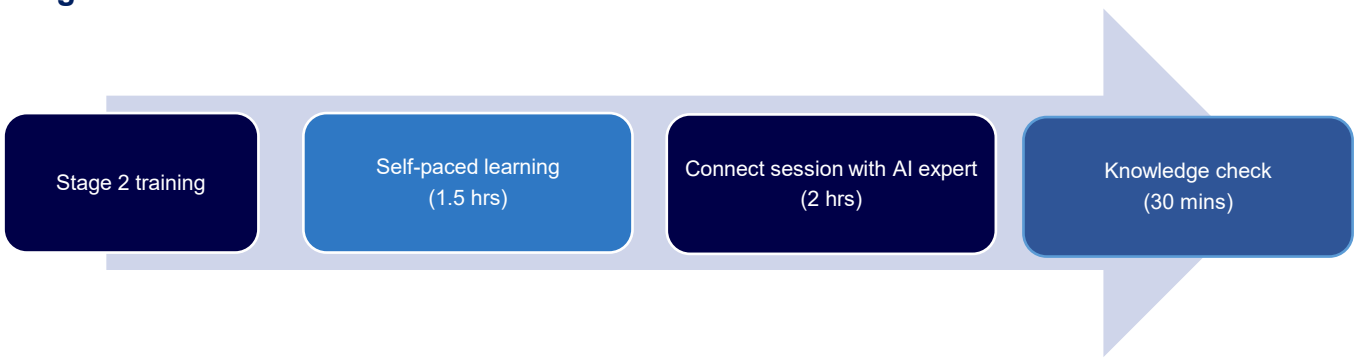
- **Learning:** GenCs will have access to curated content and resources that cover a wide range of topics related to AI, including best practices, and case studies. This learning aspect aims to deepen GenCs understanding of AI and its applications in various industries.
- **Expert connect:** GenCs will have the opportunity to connect with expert in the field of AI. This expert will provide guidance, support, and insights to help GenCs navigate their learning journey and gain valuable insights into the industry.
- **Practice sessions:** GenCs can practice the use cases provided sessions that are designed to reinforce their learning and help them apply their knowledge in real-world scenarios. These sessions will provide GenCs with hands-on experience and practical skills that are essential for success in the field of AI.

Performance outcomes

Upon completing the self-paced learning component of AI Accelerate GenCs are expected to achieve the following performance outcomes:

- GenCs will demonstrate a thorough understanding of the fundamentals of AI, including key concepts, terminology, and principles.
- GenCs will be able to apply AI concepts and techniques to solve real-world problems, demonstrating their ability to analyze, design, and implement AI solutions.
- GenCs will develop and apply critical thinking and problem-solving skills in the context of AI, enabling them to identify, analyze, and address complex challenges.
- GenCs will collaborate effectively with peers, mentors, and industry experts to achieve common goals and contribute to the advancement of AI knowledge and practice.

Program workflow



Java Programming - Overview

What is Java Programming?

Java Programming is a versatile and widely used programming language that enables developers to create robust, high-performance applications. The process of Java programming involves several stages to ensure the development of efficient and maintainable software.

Purpose

Java programming is designed to be platform-independent, allowing developers to write code that runs on any device with a Java Virtual Machine (JVM). It emphasizes object-oriented principles, making it easier to create modular, reusable, and maintainable code.

For Example, Mobile Applications: Java is the primary language for Android app development. Many popular apps like Spotify, Twitter, and Netflix are built using Java12. Enterprise Applications: Java is extensively used in enterprise environments for building large-scale systems such as banking applications, customer relationship management (CRM) tools, and enterprise resource planning (ERP) solutions.

Stages of Java Programming:

1. **Requirement Analysis**: Understand and gather the requirements for the Java application. This involves discussions with stakeholders to define the scope and objectives.

2. **Design**: Create a blueprint for the application. This includes designing the architecture, selecting design patterns, and planning the user interface and data flow.

3. **Development**: Write the Java code based on the design specifications. This stage involves coding, unit testing, and integrating different modules.

4. **Testing**: Perform various tests (unit, integration, system, and acceptance testing) to ensure the application functions correctly and meets the requirements.

5. **Deployment**: Deploy the Java application to the production environment. This may involve setting up servers, configuring environments, and ensuring the application is accessible to users.

6. **Maintenance**: Provide ongoing support and updates to the application. This includes bug fixes, performance improvements, and adding new features.



Software Requirements

Below are the software's need to be installed in system from Company Portal, before start proceeding with selenium Handson and projects like mini project and hackathon.

1. Eclipse Org Eclipse IDE 2023-12
2. Azul JDK Version 17
3. Google Project Hosting Selenium Java 4.15.0
4. Google Project Hosting Selenium Server Standalone 4.8.0
5. Apache Software Foundation, The Apache POI 5.2.4
6. Eclipse Org TestNG for Eclipse 7.4.0
7. Jenkins project Jenkins 2.251

NOTE: If genc's are facing any challenges while installation, trainer/mentor will help to install the software.

Objective to use the software:

•Developing the project
•Writing the code
•Debugging the errors
•Testing the applications

BU Mandatory - Technical learning and Handson – Core Java

NOTE: The Below topics will be covered by Trainers in Master sessions and relevant

hands-on exercises will be provided as Daily Learning Exercises (DLE) for associates to practice.

Associates must complete the provided Percipio courses along with these BU Connect Master sessions. Hands-on practice and doubt clarification can be addressed during the trainer-led Master sessions.

Associates must also complete the self-paced learning and hands-on exercises available on the Tekstac Platform in parallel with the trainer-led sessions and DLE.

Click Here to View the BU Aligned Technical Training Contents Document.

Schedule – Stage 1: Day 1 to 15

Day 1 to 15 will be focusing on Core Java.

Percipio learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Continuous Learning: Technical Hands-on through Tekstac

Post Completion of the ILT session, Trainee must Practice the Handson provided in the Tekstac Platform . In build code editor is available in this platform and Handson are automatically evaluated when the code is compiled in the editor.

Continuous Learning: Java Practice Hands-on in Eclipse

You will Parallel Learn and complete the Tekstac Platform Hands-on exercises and practice Topic wise Java Practice Handson Daily . This Exercise is set for practice only and not mandate . But completion of Tekstac Platform Handson is mandate for enablement of Assessments.

Follow these steps to practice - Topic Wise Java Practice Hands-on :

1. Software: Download the Eclipse software.

2. Practice Topic-wise Hands-on Exercises: Practice the given topic-wise hands-on exercises in your local environment using Eclipse daily post completion of Learning and platform driven Handson Exercise.
3. Submit for Evaluation: Once you have completed the exercises, submit them for evaluation.

BU Mandatory - Self Learning Contents – Core Java

Day 1

Core Java

Continuous Learning: ILT Enablement

Chapter	Chapter Name	Topic	Training Mode
Chapter-1	**Introduction to Java,Data Types and Variables**	**History & Features of Java:** OOP, platform independence, JVM, JDK, JRE	In-Person Trainer - ILT (ILT - Instructor Led Training)
		Data Types: Primitive and non-primitive types, Variable declaration, Type casting	
Chapter-2	**OOP Basics**	**Object-Oriented Principles:** Encapsulation, Inheritance, Polymorphism, Abstraction	
		Classes and Objects: Class structure, Object creation, Instance variables	
		Constructors: Default constructor, Parameterized constructor, Constructor overloading	
		HandsON : Day 1 Topics - TekStac Lab (Practice In TekStac Platform)	Daily Lab Exercise(DLE)

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Percipio.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Streamline Your Health Journey
- Water Tank Capacity
- Square Series

Continuous Learning: Java Practice Hands-on in Eclipse

Basic Syntax

- Simple Java program
- Print an integer value

- Command line arguments
- Print the input from scanner

Object Oriented Programming

- Object Oriented Programming
- Constructors
- Abstract Classes
- Interfaces
- Static Members
- Final Keyword
- Miscellaneous
- Factorial of a number using recursion
- Find the factorial of a given number
- See the output as 0, 1, 1, 2, 3
- Find the biggest number among 1,2,3,4,5,65,76,5,4,33,4,34,232,3,2323
- Read and write a file
- Print the below output
- Print the triangle of numbers

Learn and Practice

Learn Java

Refer relevant sections in this Percipio course and complete the corresponding learnings.

Day 2

Core Java

Continuous Learning: ILT Enablement

Chapter	Chapter Name	Topic	Training Mode
Chapter-3	**Inheritance,Abstraction & Encapsulation**	**Inheritance:** Single inheritance, Multilevel inheritance, super keyword	In-Person Trainer - ILT
		Polymorphism: Method overloading, Method overriding	
		Abstraction: Abstract classes, Interfaces	
		Encapsulation: Access modifiers (public, private, protected), Encapsulating data	
		HandsON : Day 2 Topics - TekStac Lab (Practice In TekStac Platform)	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Percipio.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- WordZee
- Fishing competition
- Brilliant Restaurant
- Investment Calculation

Continuous Learning: Java Practice Hands-on in Eclipse

Basic Arithmetic Operations

- Java program to swap two numbers
- Convert from Fahrenheit to Celsius
- Swap two numbers without using a third variable
- Java program to add two numbers
- Java program to find the GCD of two numbers
- Java program to find the LCM of two numbers
- Java program to print the sum of digits of a number
- Java program to find the GCD of two numbers
- Java program to find the LCM of two numbers
- Java program to print the sum of digits of a number

Learn and Practice

Learn Java

Refer relevant sections in this Percipio course and complete the corresponding learnings.

Day 3

Core Java

Continuous Learning: ILT Enablement

Chapter	Chapter Name	Topic	Training Mode
Chapter-4	**Keywords**	**this and super:** Referring current object, Accessing parent class methods	In-Person Trainer - ILT
		Static Members: Static variables, Static methods, Static blocks	
		final Keyword: Final variables, Final methods, Final classes	
Chapter-5	**Arrays,Strings,StringBuffer & StringBuilder**	**Arrays:** One-dimensional array, Multidimensional array, Array traversal and operations	
		Strings: String class, Immutability, String methods: length, substring, equals, etc.	
		HandsON : Day 3 Topics - TekStac Lab (Practice In TekStac Platform)	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.

3. Refer to and learn the same topics covered by the trainer on Percipio.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Dazzle Closet
- Mass Of Rocket Component
- Lego Builders
- Plan Smart
- Mythic Rhythms

Continuous Learning: Java Practice Hands-on in Eclipse
Control Flow Statements

- Demonstrate if…else statement
- Demonstrate nested if…else if… statement
- Demonstrate nested if…else statement
- Find odd and even numbers
- Printing the prime numbers

Learn and Practice

Learn Java
Refer relevant sections in this Percipio course and complete the corresponding learnings.

Day 4

Core Java

Continuous Learning: ILT Enablement

Chapter	Chapter Name	Topic	Training Mode
Chapter-5	**Arrays,Strings,StringBuffer & StringBuilder**	**StringBuffer & StringBuilder:** Mutability, Common methods: append, insert, delete	
Chapter-6	**Exception Handling**	**Types of Exceptions:** Checked exceptions, Unchecked exceptions	
		try-catch Block: Single try-catch, Multiple catch, finally block	
		throw and throws: Manually throwing exceptions, Propagating exceptions	In-Person Trainer - ILT
		HandsON : Day 4 Topics - TekStac Lab (Practice In TekStac Platform)	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Percipio.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Fitness Tracker
- Array Aggregate
- Product Details

Continuous Learning: Java Practice Hands-on in Eclipse

Loops

- Demonstrate for loop
- Print stars using for loop, where the number of stars printed should be equal to the row number
- Demonstrate while loop
- Print the entered number in reverse
- Demonstrate the usage of break statement inside while loop
- Print the alphabets using for loop
- Demonstrate the usage of break and continue statements inside while loop
- Demonstrate for each loop
- Print Floyd's triangle
- Print Pascal's triangle
- Print the Fibonacci series
- Printing the Multiplication table
- Find the factorial of a number
- Print the prime numbers up to a given number
- Calculate the power of a number using a while loop
- Verify whether a number is a perfect number or not
- Printing the Fibonacci series from 1 to 10
- Find the greatest of three numbers
- Print the largest two numbers from an array of given numbers

Learn and Practice

Learn Java

Refer relevant sections in this Percipio course and complete the corresponding learnings.

Additional Learning:

Technical Quizzes:

- Java Operator, Control flow statement

Day 5

Core Java

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic	Training Mode
Chapter-6	**Exception Handling**	**Custom Exceptions:** Creating custom exceptions	
Chapter-7	**Java Collections**	**Introduction to Collections:** Collection hierarchy, Collection interfaces (List, Set, Map)	
		List Interface: ArrayList, LinkedList, Vector, Stack	In-Person Trainer - ILT
		HandsON : Day 5 Topics - TekStac Lab (Practice In TekStac Platform)	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Percipio.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- College Namelist
- Babitha's App
- Alta Motors

Continuous Learning: Java Practice Hands-on in Eclipse

Method Overloading and Overriding

- Final methods be overloaded
- Static methods be overloaded
- Final methods be overridden
- Static methods be overridden
- Overload a main method in Java
- Object u have created

POLYMORPHISM & INHERITANCE

- Demonstrate method overloading
- Demonstrate method overriding
- Demonstrate inheritance
- Demonstrate the use of the 'super' keyword

Learn and Practice

Learn Java

Refer relevant sections in this Percipio course and complete the corresponding learnings.

Day 6

Core Java

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic	Training Mode
Chapter-7	**Java Collections**	**Set Interface:** HashSet, LinkedHashSet, TreeSet	In-Person Trainer - ILT
		Map Interface: HashMap, LinkedHashMap, TreeMap, Hashtable	
		Iterator: Iterator, ListIterator	
		HandsON : Day 6 Topics - TekStac Lab (Practice In TekStac Platform)	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Percipio.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Java Practice Hands-on in Eclipse

Arrays

- Find the largest number
- Remove duplicates from an array
- Find the second largest number in an array?
- Sort an array using bubble sort
- Sort an array using selection sort
- Sort an array using insertion sort
- Implement binary search
- Implement linear search
- Find the sum of elements in an array
- Find the average of elements in an array
- Find the maximum and minimum elements in an array
- Remove duplicates from an array
- Find the smallest and largest numbers in a Java Array
- Java program to swap two strings without using temp or third variable
- Demonstrate creating an array
- Duplicate elements in an array of numbers
- Demonstrate creating a multidimensional array
- Print the elements of the array in reverse
- Print alternative elements in a String array
- Find the greatest number in an integer array
- Find the least number in an integer array
- Using a for loop with single-dimensional arrays
- Using a for-each loop with single-dimensional arrays

Learn and Practice

JDBC: An Introduction to Java Database Connectivity
https://www.geeksforgeeks.org/java/establishing-jdbc-connection-in-java/

- Go through entire course.
- Implement the examples along with the author

Additional Learning:
Serial and Parallel Sorts in Java

- https://www.geeksforgeeks.org/serial-sort-vs-parallel-sort-java/

Streams and Optionals

- https://www.geeksforgeeks.org/java-8-stream-tutorial/?ref=lbp

Day 7

Core Java

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic	Training Mode
Chapter-7	**Java Collections**	**Comparable & Comparator:** Sorting with Comparable, Sorting with Comparator	
Chapter-8	**Java Thread Handling**	**Introduction to Threads:** Thread vs Process, Thread lifecycle	
		Creating Threads: Extending Thread class, Implementing Runnable interface	
		Thread Methods: sleep(), join(), start(), run()	
Chapter-9	**Java File Handling**	**Character Streams:** FileReader, FileWriter, BufferedReader, BufferedWriter	In-Person Trainer - ILT
		HandsON : Day 7 Topics - TekStac Lab (Practice In TekStac Platform)	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Percipio.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac
Mandatory Hands-on

- Student Score Info
- Vintage Books Emporium

Practice Section – Integrated Capability Test

Once completing Percipio course, Learnings, Handson related to Java skill appear for the Practice Assessment. This assessment has been set up in platform to help you to assess and apply the concepts learnt as part of Java skill.
You can take up this assessment as a practice till your qualifier assessment for practice. The purpose of this practice assessment is to prepare you for the actual assessment and enhance your information retention. Regular practice helps reinforce what you've learned, making it easier to recall during the actual assessment.

Assessment Attempts: Unlimited
Assessment Duration: 3 hrs
No No.of Questions appear in each attempt : 3

Continuous Learning: Java Practice Hands-on in Eclipse
String Handling

- Java program to compare two strings
- Java program to reverse a string
- Java program to count the number of vowels and consonants in a string
- Java program to find the frequency of characters in a string
- Java program to reverse a string
- Java program to check if two strings are anagrams
- Java program to count the number of vowels and consonants in a string
- Java program to find the frequency of characters in a string
- A string such that the first character is uppercase, the second is lowercase, and so on the numbers from a string and print only the alphabets
- Reverse the words in a sentence
- The frequency of words in a paragraph

Learn and Practice

Learn Java

Refer relevant sections in this Percipio course and complete the corresponding learnings.

Day 8

Core Java

Continuous Learning: Technical Enablement

Chapter	Chapter Name	Topic	Training Mode
Chapter-9	**Java File Handling**	**File Handling:** File class, File reading and writing (FileInputStream, FileOutputStream)	
Chapter-10	**JDBC (Java Database Connectivity)**	**JDBC API Overview:** JDBC architecture, Driver Manager, Statement and PreparedStatement	
		CRUD Operations: Insert, Update, Delete, Select Queries	
		Transactions: Commit, Rollback, Savepoints	In-Person Trainer - ILT
		HandsON : Day 8 Topics - TekStac Lab (Practice In TekStac Platform)	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Percipio.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac
Mandatory Hands-on

- Animalia
- Customer Record

Continuous Learning: Java Practice Hands-on in Eclipse
Exception Handling

- Demonstrate exception handling using try-catch
- Demonstrate the use of finally block
- Demonstrate the use of throw and throws
- Demonstrate custom exceptions
- Demonstrate the advantage of finally in Exception Handling
- Custom class which is immutable

Learn and Practice

Learn Java

Refer relevant sections in this Percipio course and complete the corresponding learnings.

Day 9

Core Java

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic	Training Mode
Chapter-11	**Java 8 Features**	**Lambda Expressions:** Lambda syntax, Functional interfaces	
		Stream API: Stream operations (filter, map, reduce)	
		Optional Class: Avoiding NullPointerException with Optional	
Chapter-12	**Generics & Annotations**	**Generics:** Generic classes, methods & wildcards	In-Person Trainer - ILT
		HandsON : Day 9 Topics - TekStac Lab (Practice In TekStac Platform)	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Percipio.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac
Mandatory Hands-on

- Fit Freak
- Employee Details

Continuous Learning: Java Practice Hands-on in Eclipse
Collections Framework

- Java program to demonstrate the use of collections (ArrayList)
- Java program to demonstrate the use of collections (HashMap)
- Java program to demonstrate the use of collections (HashSet)
- Java program to demonstrate the use of collections (LinkedList)
- Java program to demonstrate the use of collections (TreeSet)
- Java program to demonstrate the use of collections (PriorityQueue)
- Java program to demonstrate the use of collections (Stack)
- Java program to demonstrate the use of collections (Vector)

Learn and Practice

Learn Java

Refer relevant sections in this Percipio course and complete the corresponding learnings.

Day 10

Core Java

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic	Training Mode
Chapter-12	**Generics & Annotations**	**Annotations:** Built-in annotations (@Override, @Deprecated) & Custom annotations	
Chapter-13	**Design Patterns**	**Design Patterns:** Singleton, Factory, Observer, Strategy patterns	In-Person Trainer - ILT
		HandsON : Day 10 Topics - TekStac Lab (Practice In TekStac Platform)	DLE & BU Connects

Continuous Learning: Technical Enablement through Self-learning

After completing each chapter of the ILT (Instructor-Led Training), trainees should:

1. Practice the mandatory hands-on exercises on the Tekstac platform.
2. Work on the provided hands-on exercises in Eclipse.
3. Refer to and learn the same topics covered by the trainer on Percipio.

This approach ensures comprehensive understanding and reinforcement of the topics.

Continuous Learning: Technical Hands-on through Tekstac
Mandatory Hands-on

- Booking System
- Copy Names

Continuous Learning: Java Practice Hands-on in Eclipse
Collections Framework

- Java program to demonstrate the use of collections (Deque)
- Java program to demonstrate the use of collections (LinkedHashMap)
- Java program to demonstrate the use of collections (LinkedHashSet)
- Find duplicates in a list
- Two HashMaps for equality
- Creating an ArrayList
- Creating a LinkedList
- Creating a HashSet
- Creating a LinkedHashSet

- Creating a TreeSet
- Creating a PriorityQueue
- Creating a HashMap using the Map interface
- Creating a LinkedHashMap
- Creating a TreeMap

Learn and Practice

Learn Java

Refer relevant sections in this Percipio course and complete the corresponding learnings.

Continuous Learning: Technical Hands-on

Additional Learning:
Technical Quizzes:

- Applying Object Oriented Concepts in java

Day 11 & 12

Core Java and JDBC

Continuous Learning: Technical Enablement through Tekstac

Mandatory Learning

Java 8 Features - Streams and Optionals. Asynchronous and Parallel Programming in Java 8.
Go through web pages for learning below specific topics

Asynchronous and Parallel Programming
Java User Input (Scanner)

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Mystic Line
- Area And Volume
- The Perfect Password
- Library Management

Learn and Practice

Learn Java

Refer relevant sections in this Percipio course and complete the corresponding learnings.

Additional Learning:
Technical Quizzes:

- Quiz 3 - Collections Framework
- Quiz 4 - Advanced Java Concepts

Once completing course, Learnings, all Handson & Practice assessments related to Java skill appear for the Java -Code Challenge Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Assess-Type-1: Code Challenge – Java - Group 1
Assess-Type-1: Code Challenge – Java - Group 2

Assessment Duration: 2 hrs.
Assessment Attempts: 3
No.of Questions appear in each attempt: 1

Day 13 to 15

Core Java

Continuous Learning: Technical Enablement

Learn and Practice

Aspire Journeys: Automated Software Testing with Java Selenium Track 1: Selenium Testing

JDBC: An Introduction to Java Database Connectivity
https://www.geeksforgeeks.org/java/establishing-jdbc-connection-in-java/

- Rehearse the sections based on need

Once completing course, Learnings, all Handson & Practice assessments related to Java skill appear for the Java - ICT Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

GenC - QAQE-TECHNICAL-JAVA TRACK - CORE JAVA SKILL WAR - SKILL BASED ASSESSMENT [101-BASICS] -LAB_ATKJE072

Assessment Duration: 4 hrs.
Assessment Attempts: 3
No.of Questions appear in each attempt : 1

Schedule – Stage 1: Day 16 to 21

Day 16

SQL

Day 16 to 21 will be focusing on SQL Language.

Percipio learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Continuous Learning: Technical Enablement

Learn the basics of ANSI SQL with MySQL

Refer required sections in this course and complete the corresponding learnings.

Go through web pages for learning below specific topics

RANK Function

Continuous Learning: Technical Hands-on
Mandatory Hands-on

- Create Borrowing Table
- Alter Table User
- Alter Table Book
- Remove Table
- Insert User Table
- Update Record
- Delete Record

Day 17

SQL

Continuous Learning: Technical Enablement

Learn the basics of ANSI SQL with MySQL

Refer required sections in this course and complete the corresponding learnings.

Continuous Learning: Technical Hands-on
Mandatory Hands-on

- Diverse Selection
- Email Providers
- Book Event
- Upcoming Fair
- Routine Checks
- Security Measure
- May Birthday
- Age Demographics
- Author Detail

Additional Learning:

Technical Quizzes:
Querying Database - Operators, Aggregate, String, Date Functions

MOCK Stage1 Assessment

Once completing Percipio course, Learnings, all Handson & Practice assessments related to Java skill appear for the **Mock Stage1 Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform.

Assessment Duration: 3 hrs.
Assessment Attempts: 3
No.of Questions: 3

Day 18

SQL

Continuous Learning: Technical Enablement

Learn the basics of ANSI SQL with MySQL

Refer required sections in this course and complete the corresponding learnings.

Continuous Learning: Technical Hands-on

Mandatory Hands-on

- Price Analysis
- Updated Email
- Lended Books
- The Chronicles of Authors and Their Tales
- Date Format
- Exploring Users' Literary Journeys
- Active Borrowings with User and Book Details
- Same Authors
- User Borrowing Summary
- Maximum Fine Per User
- Top Users
- User Count by Title and Author

Additional Learning:

Technical Quizzes:
SQL basics with DML and DDL Statements

Once completing Percipio course, Learnings, all Handson & Practice assessments

related to Web UI-SQL skill appear for the SQL -Code Challenge Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge
Assess-Type-1: Code Challenge - DDL, DML & Select (QEA)

Assessment Duration: 2 hrs.
Assessment Attempts: 3
No.of Questions appear in each attempt : 1

Day 19

XML and JSON

Continuous Learning: Technical Enablement

Learn the basics of XML and JSON

Once completing Percipio course, Learnings, all Handson & Practice assessments related to Web UI-SQL skill appear for the SQL -Code Challenge Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge
Assess-Type-1: Code Challenge - JOINS and SUBQUERIES (QEA)
Assess-Type-1: Code Challenge - Scalar, Aggregate (QEA)

Assessment Duration: 2 hrs.
Assessment Attempts: 3
No.of Questions appear in each attempt : 1

Day 20 & Day 21

Once completing Percipio course, Learnings, all Handson & Practice assessments related to Java skill appear for the **Stage 1 Assessment**. This will help you to assess and apply the concepts of the skill learnt in the platform. Question Pattern - 3 SBA Java Questions approximately spend around 1 hr. for each SBA Question. Score acquiring in 1 st instance will be getting into consideration. There are no more attempts provided further.

Stage1 Assessment

Assessment Duration: 3 hrs.
No. of Questions: 3

Learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Day 22

Agile Functional Testing

Topic #	Topic Name	Topic Objective	References
Module 1	Agile Software Testing Concepts	Understanding SDLC and STLC	Theory and Foundations
		Distinguishing between Project and Product in testing context	
		Exploring V-Model and the roles of QA, QC, and QE	
		Differentiating between White Box and Black Box Testing	
		Delving into Static Testing and Dynamic Testing	
		Conceptualizing Verification and Validation	
		System Testing Types (GUI Testing)	
		System Testing Types (Functional vs. Non-Functional Testing)	
		Test Design Techniques and strategies	Includes Re-Testing, Regression Testing, Exploratory Testing, Adhoc, Sanity, Smoke Testing, and End-To-End Testing.
		Understanding STLC phases (Use case, Test scenario, Test case)	
		Understanding STLC phases (Test Environment, Execution, Defect Reporting)	
		Understanding STLC phases (Test Closure, and Test Metrics)	
Module 2	Agile Software Testing Project	Introduction to the Live Project	Practical Application
		Extracting Functional Requirements from FRS	Requirement Analysis
		Creating comprehensive Test Scenarios and Test Cases	Writing Test Artifacts
		Execution of Test Cases	
		Reporting and Tracking Bugs	Defect Management
		Test Sign-off	Completion Phase
Module 3	Agile/Scrum Process	Overview of Agile and Scrum methodologies	Agile Fundamentals
		Understanding Scrum Teams, Sprints, and User Stories	
		Estimating User Stories and defining 'Definition of Done' and 'Definition of Ready'	
		Exploring Sprint Activities (Planning, Refinement, Review, Retrospective)	

Module 4	JIRA Tool Implementation	Installation and Configuration of JIRA	Hands-on Tool Setup
		Project creation in Jira and user management	
		Creating Backlog, Epic, and User Stories in JIRA	Requirements in JIRA
		Sprint creation and management	
		Understanding the Sprint life cycle in JIRA	
		Bug creation in Jira	Defect Reporting in JIRA
		Writing test cases in JIRA	Test Case Management
		Creating Test Cycles and executing/updating Test Cases in Jira	Test Execution in JIRA

Continuous Learning: Technical Enablement

Learn the basics of Agile Fundamentals and Software Testing Life Cycle

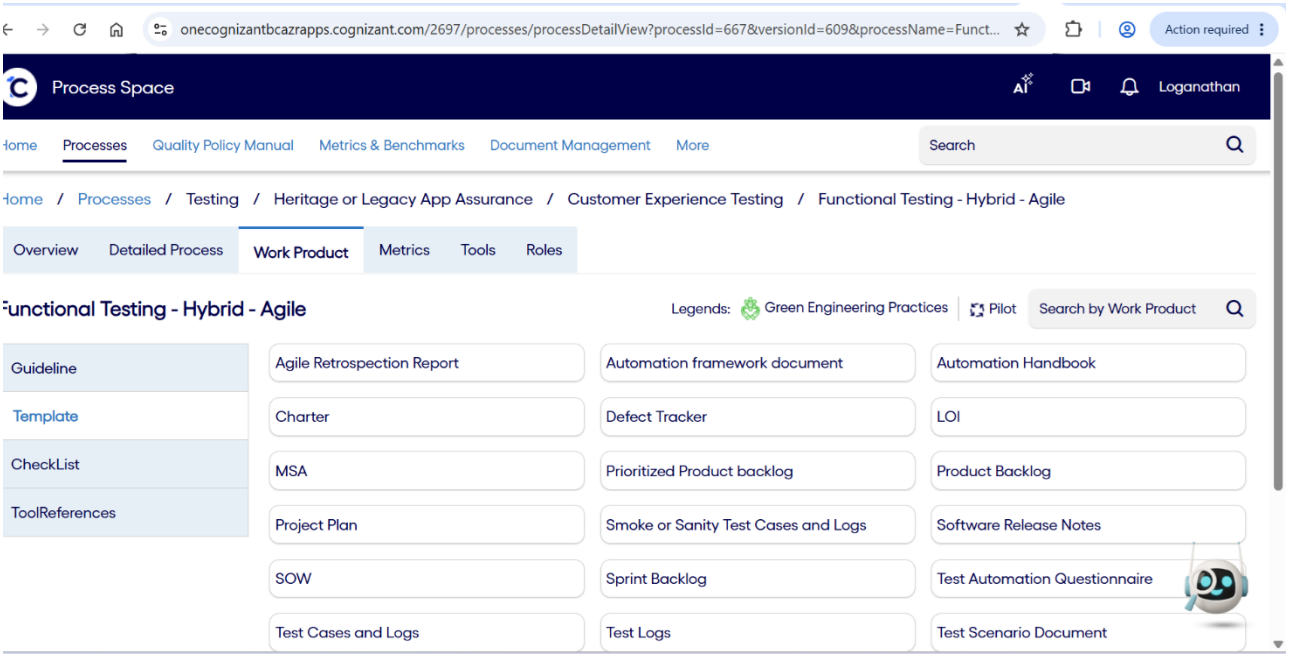
Software Testing

Learn all the sections in this Percipio course to gain deep understanding about below modules

Mandatory Templates for Hands-on Practice

******* All trainees must use the standard Cognizant functional testing templates (e.g., Test Strategy, Defect Tracker) for every activity, as provided in the link below.**
Use of any self-created or non-standard templates by trainers or trainees will not be accepted and will not be considered during evaluations. During Interim or Final interviews, evaluators may request a demo and only work completed using the prescribed templates will be eligible *****

https://onecognizantbcazrapps.cognizant.com/2697/processes/processDetailView?processId=667&versionId=609&processName=Functional%20Testing%20-%20Hybrid%20-%20Agile&archetype=Heritage%20or%20Legacy%20App%20Assurance&workCategory=Customer%20Experience%20Testing&cluster=Testing



Module : Software Testing Concepts

- Understanding Software Development Life Cycle (SDLC) and Software Testing Life Cycle (STLC)
- Distinguishing between Project and Product in the context of testing
- Exploring V-Model and the roles of QA (Quality Assurance), QC (Quality Control), and QE (Quality Engineering)
- Differentiating between White Box and Black Box Testing
- Delving into Static Testing and Dynamic Testing
- Conceptualizing Verification and Validation in software testing
- System Testing Types, including GUI Testing and Functional vs. Non-Functional Testing
- Test Design Techniques and strategies like Re-Testing, Regression Testing, Exploratory Testing, Adhoc Testing, Sanity, Smoke Testing, and End-To-End Testing
- Understanding Software Testing Life Cycle (STLC) phases, including Use case, Test scenario, Test case, Test Environment, Execution, Defect Reporting, Test Closure, and Test Metrics.

Module : Software Testing Project

- Introduction to project
- Extracting Functional Requirements from Functional Requirement Specifications (FRS)
- Creating comprehensive Test Scenarios and Test Cases
- Execution of Test Cases
- Reporting and Tracking Bugs
- Test Sign-off

Phase : Agile Testing + Jira Tool

Agile/Scrum Process:

- Overview of Agile and Scrum methodologies
- Understanding Scrum Teams, Sprints, and User Stories
- Estimating User Stories and defining 'Definition of Done' and 'Definition of Ready'
- Exploring Sprint Activities such as Sprint Planning, Backlog Refinement, Sprint Review, and Sprint Retrospective

Jira Tool:

- Installation and Configuration of JIRA
- Project creation in Jira and user management
- Creating Backlog, Epic, and User Stories in JIRA
- Sprint creation and management
- Understanding the Sprint life cycle in JIRA
- Bug creation in Jira
- Writing test cases in JIRA with Zephyr plugin
- Creating Test Cycles and executing/updating Test Cases in Jira

Continuous Learning: Technical Hands-on

Mandatory Hands-on

- The Importance of QEA and Career Scope
- The Scope Debate: Quality Assurance vs. Software Development
- Trainer-Guided Integrated Project: Activity Readiness
- Trainer-Guided Integrated Project: Configured Jira Project & Backlog Management
- Trainer-Guided Integrated Project: Test Case Development in Jira (using Zephyr/similar plugin)
- Trainer-Guided Integrated Project: Test Execution & Status Reporting Artifacts
- Trainer-Guided Integrated Project: User story document
- Trainer-Guided Integrated Project: Retrospection Report
- Trainer-Guided Integrated Project: Sprint Backlog
- Placeholder to submit other Functional agile deliverables

    - ✓ **You should have introdcued more than one sprints**
    - ✓ **Ensure you have in-sprint automation concepts introduced as part of project**
    - ✓ **Ensure you have created Regression suite and smoke suite**
    - ✓ **Ensure your regression suite consists of minimum 25 test cases and test scripts**
    - ✓ **Ensure your smoke suite covers end to end business functionality**

Additional Learning:
Technical Quizzes:
Functional Testing

Day 26

Functional Testing

Once completing Percipio course, Learnings, all Handson & Practice assessments related to Functional Testing skill appear for the Functional Testing -ICT Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform. Attempts allowed to practice the assessment is 3.

Assessment Duration: 4 hrs
Assessment Attempts: 3
No.of Questions appear in each attempt : 1

Functional Testing -Assess Type 2 Skill Based Assessment.

- Read the requirements carefully of the Assessment Question.
- Write the Defect Description for the Test Case in Excel Sheet
- Submit the Test deliverables

GenAI (Day 24,Day 26)

Continuous Learning: Technical Enablement

AI Accelerate

By this course you will complete GenAI self-pace learning , AI expert connect session and AI Knowledge check assessment.

Program completion criteria

Everyone must register for the following e-learning course on **C-Learn** and complete a KBA assessment on Moodle to successfully finish this program.

Online learning: C-Learn



Activity Code: **ELRNG01863**

Fundamentals of Generative AI [101-Basics]

This AI course is designed to equip learners with:

- The foundational knowledge and skills required to harness the power of Generative AI.
- The ability to identify opportunities for innovation and implementation of AI within their organizations.
- The skills to drive organizations toward a future of enhanced creativity and competitive advantage using AI techniques.

Day 27 to 31 will be focusing on Selenium configuration and WebDriver Basics.

Percipio learnings are recommended in the Platform to understand the fundamental concepts.

Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Integrated Project Deliverable: Agile and Functional Testing handson

As the selenium learning starts, the project details(By trainer) will be given for the learners so that you can parallel keep doing the project activities along with the rest of the learnings.

Please proceed with the automation implementation for the functional requirements provided in the previous milestone ("Trainer Given Handson").

Instructions: As you progress through this milestone's learning objectives (Selenium Architecture, Environment Setup, WebDriver, Locators, and Customized XPATH), incrementally integrate these newly acquired skills into your existing automation project.

Objective: The goal is to build a foundational, executable set of automated test scripts that validate the functionality defined in the agile documents you previously prepared.

The given Project are expected to be completed on or before the end week of the program.

Project Evaluation will be based on:

- Source Code
- Functionality Completion, Usage of Features, Code Quality
- Demo of Output

Business Aligned Project (Mini Project/ Hackathon)

As the selenium learning starts, the project details (Mini project and the Hackathon) will be given for the learners so that they can parallel keep doing the project activities along with the rest of the learnings.

• Mini project in selenium is a great way to deepen your understanding, showcase your skills, and prepare for real-world automation tasks.
• It allows to apply theoretical knowledge of Selenium and web automation in a practical, real-world scenario. Working on a project helps you gain hands-on experience with Selenium WebDriver, locators, and other essential components.
• Building a mini project helps you understand the design and implementation of automation frameworks, such as Page Object Model (POM) and Data-Driven Testing.
• You can integrate Selenium with other tools like TestNG for test management and reporting.

Mini Project & Hackathon – Objective

• CI/CD Integration: Integrating your Selenium tests with CI/CD tools like Jenkins helps automate the testing process as part of the build pipeline.
• Automated Testing: Ensures that your application is tested automatically with every code change, improving software quality.
• Reporting and Logging: Generate detailed test reports that provide insights into test execution, pass/fail status, and error details.
• Logging: Implement logging to capture detailed information about test execution, which is useful for debugging and analysis.
• Reusable Code: Writing reusable and maintainable code is a key aspect of any project, and a mini project helps you practice this.
• Collaboration: It encourages collaboration and teamwork, as you can divide tasks among team members and integrate their work using Selenium.
• Exception Handling: You'll learn to handle exceptions and errors that occur during test execution, making your scripts more robust.
• Debugging Skills: Debugging skills are enhanced as you troubleshoot issues in your automation scripts etc.

The given Mini Project and the Hackathon are expected to be completed on or before the end week of the program.

Project Evaluation will be based on:

- Source Code
- Functionality Completion, Usage of Features, Code Quality
- Demo of Output

Note:

Mini Project is an individual team member activity. Project Requirement is available in the platform. Candidates must complete the requirement and submit the solution for evaluations by BU SME.

Hackathon is an group activity. Project requirement is available in the platform. Candidates must complete the requirement and submit the solution for evaluations by BU SME.

Selenium Automation for Realtime Project				
Section #	Module / Topic	Detailed Lecture / Sub-Topic	Concept/Tool	Notes
1	Setting the Project Foundation	Automation Project Setup and Environment Configuration	Java, Maven/Gradle	Dependencies and Build Automation.
		Introduction to Test Design Patterns	Page Object Model (POM)	Structure, benefits, and implementation.
		Setting up Logging and Reporting Tools	Log4j, Extent Reports	Generating detailed, professional reports.
2	Advanced WebDriver Concepts	Synchronization Techniques	Explicit & Fluent Waits	Handling dynamic elements and improving stability.
		Handling Complex Web Elements	Action Class, JavaScript Executor	Mouse and Keyboard interactions, scrolling.
		Handling Popups, Windows, and Frames	WebDriver	Switching contexts and managing alerts.
		Capturing and Analyzing Failure Points	Taking and Organizing Screenshots	Capturing evidence upon test failure.
3	Designing the Automation Framework	Hybrid Framework Design	POM + Data-Driven + Keyword-Driven	Combining best practices into one framework.
		Data Management for Live Projects	Data-Driven Testing	Reading test data from Excel, JSON, or databases.
		Implementing Robust Exception Handling	Java	Writing code that handles unexpected errors gracefully.
		Writing Reusable Utility Methods	Custom Libraries	Creating helper functions for common tasks (e.g., file handling).
4	Test Execution & Reporting	Parallel Test Execution	TestNG/Selenium Grid	Running multiple tests simultaneously to save time.
		Grouping and Prioritizing Tests	TestNG Annotations	Running Smoke, Sanity, or Regression Suites.
		Finalizing and Interpreting Execution Reports	Extent/Allure Reports	Analyzing results for pass/fail metrics.
5	Real-Time DevOps Integration	Integrating Framework with Version Control	GIT/GitHub	Pushing and managing the project code.
		Scheduling and Running Tests in CI	Jenkins	Setting up jobs for automated nightly/daily runs.

Day 27

Automation Concepts, configuration, WebDriver Basics

Click Here to View the BU Mandatory Technical Learning and Handson Document.

Continuous Learning: ILT Technical Enablement:
Learn the selenium automation topics from the below course

Aspire Journeys:Automated Software Testing with Java Selenium

Refer respective sections from this course and complete the corresponding Subtopics

Chapter	Chapter Name	Topic
Chapter 1	**Selenium Basics and overview**	Introduction to Browser Automation
		What is Selenium and Overview
		History and evolution of Selenium
		Selenium Components (WebDriver vs Selenium IDE vs Selenium Grid)
		Supported programming languages (Java, Python, C#, etc.)
		Cross-browser and cross-platform compatibility.
		Advantages and limitations of Selenium

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Refer to and learn the same topics covered by the trainer on Percipio.
2. Practice the Tekstac Platform Handson.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Find Element by Id
- Find Element by Name

Note: You are requested to complete below Integrated Project Deliverable on or before your day 31

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Automation Feasibility Report & Strategy
- Trainer-Guided Integrated Project: Configured Automation Environment & Project Structure
- Trainer-Guided Integrated Project: Automated Test Scripts (Individual Functional Scenarios)
- Trainer-Guided Integrated Project: Test Execution Reports and Status
- Trainer-Guided Integrated Project: Code Review and Feedback Session Artifacts

Day 28

Automation Concepts, configuration, WebDriver Basics

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 2	**Selenium Architeture**	What is Selenium Webdriver: 1) WebDriver is one of the component in selenium. 2) WebDriver is a java interface. 3) WebDriver is an API(Application Programming interface)
		Understanding the WebDriver Architecture / API methods. WebDriver, RemoteWebDriver, ChromeDriver , FirefoxDriver, EdgeDriver etc..
Chapter 3	**Setting up Selenium environment**	Installing Java Development Kit (JDK) and Eclipse IDE
		Downloading Selenium WebDriver
		Configuring Selenium WebDriver with Eclipse

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning)hands-on exercises provided in the local environment.
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 2_Handson
- Chapter 2_PPT Learning Ref.Content
- Chapter 2_Video Learning Ref.Content

- Chapter 3_Handson
- Chapter 3_PPT Learning Ref.Content
- Chapter 3_Video Learning Ref.Content

Learn the basics of WebDriver Basics:

Aspire Journeys:Automated Software Testing with Java Selenium

Refer respective sections from this course and complete the corresponding Subtopics

Day 29

Automation Concepts, configuration, WebDriver Basics

Continuous Learning: Technical Enablement

Chapter	Chapter Name	Topic
Chapter 4	**Selenium WebDriver**	Locating GUI Elements in Application Under Test (AUT)
		How to identify the locators in web page
		First Selenium Webdriver Script
		Launching Browsers in Selenium

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment Submit the solutions for evaluation.
3. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Find options in dropdown
- Find Radio Element

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 4_Handson
- Chapter 4_PPT Learning Ref.Content
- Chapter 4_Video Learning Ref.Content

Day 30

Automation Concepts, configuration, WebDriver Basics

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 5	**Locating various Web elements**	Find Element by Id,Name,Link Text,Partial Link Text,Class Name,CSS (Tag , ID , Class , Attribute , etc..)
		Handling Shadow DOM in Selenium

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Find Element by Tag Name
- Handling Check Box
- Find Element by Class Name
- LinkText and PartialLinkText

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 5_Handson
- Chapter 5_PPT Learning Ref.Content
- Chapter 5_Video Learning Ref.Content[Part-1]
- Chapter 5_Video Lear/ning Ref.Content[Part-2]
- Chapter 5_Video Learning Ref.Content[Part-3]

Additional Learning:

Technical Quizzes:
Quiz 1 - Automation Concepts, Selenium configuration, WebDriver Basics

Day 31

Automation Concepts, configuration, WebDriver Basics

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 5	**Locating various Web elements**	Relative Locators and Chained Locators in Selenium
		Locating By Xpath (Single Slash, Double Slash , contains(), starts_with() , text() , Last())

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Practice the Assessment.
5. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

Once completing Percipio course, Learnings, all Handson & Practice assessments related to Selenium-Web Driver skill appear for the Code Challenge Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge
Assess-Type-1:Code Challenge - Automation Concepts, Selenium configuration, WebDriver Basics

Assessment Duration: 2 hrs.
Assessment Attempts: 3
No.of Questions appear in each attempt : 1

Day 32

Selenium Automation Techniques, Dynamic XPath

Click Here to View the BU Mandatory Technical Learning and Handson Document.

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 5	**Locating various Web elements**	Find Element by Id,Name,Link Text,Partial Link Text,Class Name,CSS (Tag , ID , Class , Attribute , etc..)
		Handling Shadow DOM in Selenium
		Relative Locators and Chained Locators in Selenium
		Locating By Xpath (Single Slash, Double Slash , contains(), starts_with() , text() , Last())
		Find options in dropdown. Handle dropdown without using Select Class
		Find Radio Element
		Find Element by Tag Name
Chapter 6	**Desired capabilities**	Desired Capabilities in Selenium WebDriver

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- CSSLocator
- AbsoluteXpathLocator

- xPathAncestor
- Working with Alerts
- Work with Javascript Excecutor

Note: You are requested to complete below Integrated Project Deliverable on or before your day 36

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Enhanced Automation Test Scripts
- Trainer-Guided Integrated Project: Comprehensive Error Handling and Reporting
- Trainer-Guided Integrated Project: Utility Class for Advanced Functions
- Trainer-Guided Integrated Project: Code Review & Refactoring Artifacts

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 6_Handson
- Chapter 6_PPT Learning Ref.Content
- Chapter 6_Video Learning Ref.Content

Project – Mini Project Case Study

Overall Duration: Should start your mini project parallel with the Stage 2 Selenium Automation learnings.

The outcomes of doing **Mini Project** are:

- Enables learners to know on the environment setup
- Any web application is taken, and learner try to automate given scenario using Selenium APIs
- Exhibits learner skills on automation of real time applications for smaller requirement.

Day 33

Selenium Automation Techniques, Dynamic XPath

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 7	**selenium webDriver commands and Web elements interactions**	Selenium WebDriver- Commands
		Text fields and buttons
		Find Element VS FindElements in Selenium WebDriver - check order
		Handling Canvas Elements in Selenium
		Using JavaScript Executor in Selenium

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Handling RegEx Selenium
- Drag and Drop with Slider and Actions and wait
- Selenium – ScreenShot
- Selenium - DatePicker(Actions Class)
- FindElements

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 7_Handson
- Chapter 7_PPT Learning Ref.Content
- Chapter 7_Video Learning Ref.Content

Learn the basics of Automation Techniques and Dynamic Xpath

Project– Mini Project Case Study

- Integrate your mini project

Stage : GenAI Code Assist (GitHub Copilot)

Learning and practicing the below provided GenAI topics will equip you with essential, modern skills in QA Automation, preparing you for an evolving tech landscape

- **GenAI Foundation**

  This section provides the foundational knowledge of how GenAI works and the critical importance of using it ethically and responsibly.

  - Fundamentals of Generative AI [101-Basics]
  - Generative AI Quick Assessment for Elearning Quiz [101-Basics]
  - Introduction to Responsible AI
  - Responsible AI-Deep Dive [101- BASICS]

- **Prompt Engineering:** The ability to communicate effectively with AI models is a core competency for maximizing their utility. This skill directly translates to generating higher quality code suggestions, more relevant test cases, and accurate documentation, saving significant time and effort.

  - Prompt Engineering Foundation - PY0009562

- **GitHub Copilot:** This section focuses on integrating AI assistance directly into the daily workflow. Learning to use GitHub Copilot effectively helps freshers write code faster, reduce bugs through AI-driven suggestions, and understand best practices through examples provided by the AI.

  GitHub Copilot Fundamentals

Usecase and Handbook & Comprehensive GenAI Hands-On for Automation Developers: These exercises provide practical experience in optimizing AI outputs, managing token limitations, and generating functional, high-quality automation code, making the learning directly applicable to their job roles.

Day 34

Selenium Automation Techniques, Dynamic XPath

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 8	**WebDriver Actions**	Mouse and keyboard actions
		Drag and drop
		Handling frames and windows
		mouse hover (tool tip) in Selenium
		Handling tabs & cookies

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Handling ToolTip
- Working with different XPath conditions
- PurchaseApp iframe
- Multiple Window
- Selenium - Implicit Wait
- Selenium - Fluent Wait

Continuous Learning: Technical Hands-on

Best Practices- Real Time Technical Hands-on

- Desired Capabilities
- Shadow_DOM_Scenario1
- Shadow_DOM_Scenario2
- Relative Locators and Chained Locators in Selenium

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 8_Handson
- Chapter 8_PPT Learning Ref.Content
- Chapter 8_Video Learning Ref.Content

Project – Mini Project Case Study

- Integrate the mini project.

Stage : GenAI Code Assist (GitHub Copilot)

Learning and practicing the below provided GenAI topics will equip you with essential, modern skills in QA Automation, preparing you for an evolving tech landscape

- **GenAI Foundation**

  This section provides the foundational knowledge of how GenAI works and the critical importance of using it ethically and responsibly.

  - Fundamentals of Generative AI [101-Basics]
  - Generative AI Quick Assessment for Elearning Quiz [101-Basics]
  - Introduction to Responsible AI
  - Responsible AI-Deep Dive [101- BASICS]
- **Prompt Engineering:** The ability to communicate effectively with AI models is a core competency for maximizing their utility. This skill directly translates to generating higher quality code suggestions, more relevant test cases, and accurate documentation, saving significant time and effort.
  - Prompt Engineering Foundation - PY0009562

- **GitHub Copilot:** This section focuses on integrating AI assistance directly into the daily workflow. Learning to use GitHub Copilot effectively helps freshers write code faster, reduce bugs through AI-driven suggestions, and understand best practices through examples provided by the AI.

Usecase and Handbook & Comprehensive GenAI Hands-On for Automation Developers: These exercises provide practical experience in optimizing AI outputs, managing token limitations, and generating functional, high-quality automation code, making the learning directly applicable to their job roles.

Day 35

Selenium Automation Techniques, Dynamic XPath

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 9	**Multiple windows & frames**	Switching between windows
		Switching to frames & iFrames
	WebDriver Waits	Implicit waits
		Explicit waits
		Fluent waits

Chapter	Chapter Name	Topic
Chapter 10	**Selenium Automation Techniques**	Checkbox Handling and Form Submission
		Locating Web Element Attributes using CSS Selector
		Locating Web Element Attributes using Absolute XPath Locator
		Identifying Ancestors of Web Elements
		Working with alerts
		Working with Java script executor
		Working with Regular expression
		Working with Drag and Drop and Slider
Chapter 11	**Advanced WebDriver techniques**	Selenium - ScreenShot
		Handling JavaScript/window Alerts
		Headless browsers and drivers in Selenium
		Handling Dynamic Elements in Selenium
		1) get methods , 2) conditional methods, 3) browser methods, 4) navigational methods, 5) wait methods
		How to Find All/Broken links using Selenium Webdriver

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Take Up the Assessment.
5. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 9_Handson
- Chapter 9_PPT Learning Ref.Content
- Chapter 9_Video Learning Ref.Content

- Chapter 10_Handson
- Chapter 10_PPT Learning Ref.Content
- Chapter 10_Video Learning Ref.Content

- Chapter 11_Handson
- Chapter 11_PPT Learning Ref.Content
- Chapter 11_Video Learning Ref.Content

Continuous Learning: Technical Hands-on through Tekstac

Best Practices- Real Time Technical Hands-on

- Relative Locators and Chained Locators in Selenium_Scenario 1
- Relative Locators and Chained Locators in Selenium_Scenario 2
- Relative Locators and Chained Locators in Selenium_Scenario 3
- Relative Locators and Chained Locators in Selenium_Scenario 4

Additional Learning:

Technical Quizzes:
Quiz 1 - Selenium Automation Techniques, Dynamic Xpath

Project– Mini Project Case Study

- Integrate your mini project

Day 36

Selenium Automation Techniques, Dynamic XPath

Once completing Percipio course, Learnings, all Handson & Practice assessments related to Selenium – Xpath skill appear for the Code Challenge Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge
Assess-Type-1:Code Challenge - Selenium Automation Techniques, Dynamic Xpath

Assessment Duration: 2 hrs.
Assessment Attempts: 3
No.of Questions appear in each attempt : 1

Project – Mini Project Case Study

- Integrate your mini project

Day 37

You will be focusing on Selenium WebDriver with POM and ApachePOI

Percipio learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Selenium Webdriver With POM and ApachePOI

Click Here to View the BU Mandatory Technical Learning and Handson Document.

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 12	**Calendar using action**	Selenium - DatePicker(Actions Class)
	Apache POI	Apache POI - Read Operation : Excel Read Operation
		Apache POI - Write Operation: Excel Write Operation
		Applying POI POM- Flight reservation

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Excel Read Operation
- Excel Write Operation
- Applying POI - Flight reservation

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 12_Handson
- Chapter 12_PPT Learning Ref.Content
- Chapter 12_Video Learning Ref.Content

Learn the basics of WebDriver with POM and ApachePOI

Note: You are requested to complete below Integrated Project Deliverable on or before your day 41

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Refactored Automation Framework (Page Object Model Implementation)
- Trainer-Guided Integrated Project: Data Management Strategy & Implementation (Apache POI)
- Trainer-Guided Integrated Project: Execution Confirmation and Custom Logging Artifacts
- Trainer-Guided Integrated Project: Updated Agile Artifacts & Traceability (Living Documentation)

Project – Mini Project Case Study

- Integrate the mini project.

Day 38

Selenium Webdriver With POM and ApachePOI

Learn the basics of WebDriver with POM and ApachePOI

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- POM with Page Factory Model

Continuous Learning: Technical Hands-on

Best Practices- Real Time Technical Hands-on

- Mouse_Keyboard_Actions_Cookie_Scenario 1
- Mouse_Keyboard_Actions_Cookie_Scenario 2
- Mouse_Keyboard_Actions_Cookie_Scenario 3
- Mouse_Keyboard_Actions_Cookie_Scenario 4
- Mouse_Keyboard_Actions_Cookie_Scenario 5
- Mouse_Keyboard_Actions_Cookie_Scenario 6
- Mouse_Keyboard_Actions_Cookie_Scenario 7

Project – Mini Project Case Study

- Integrate the mini project.

Day 39

Selenium Webdriver With POM and ApachePOI

Continuous Learning: Technical Enablement

Learn the basics of WebDriver with POM and ApachePOI

Once completing course, Learnings, all Handson & Practice assessments related to Selenium – Apache POI skill appear for the Code Challenge Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge
Assess-Type-1:Code Challenge - Selenium Webdriver with POM and ApachePOI

Assessment Duration: 2 hrs.
Assessment Attempts: 3
No.of Questions appear in each attempt : 1

Project – Mini Project Case Study

- Integrate the mini project.

Day 40

Automation Testing - Selenium with TestNG

You will be focusing on Selenium with TestNG

Percipio learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Click Here to View the BU Mandatory Technical Learning and Handson Document.

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 13	**TestNG Concepts**	TestNG Assertions
		Introduction to TestNG
		Creating TestNG test cases
		TestNG annotations and assertions
		Data-Driven Testing in TestNG
		Running Tests in Parallel using TestNG

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- TestNG Assertions
- TestNG with two test classes
- Contact Book- POM structure
- TestNG Annotation With Priority
- TestNG Dependency

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 13_Handson
- Chapter 13_PPT Learning Ref.Content
- Chapter 13_Video Learning Ref.Content

Learn the basics of Selenium with TestNG

Note: You are requested to complete below Integrated Project Deliverable on or before your day 43

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

Mandatory Hands-on

- Trainer-Guided Integrated Project: Integrated & Structured Test Suite using TestNG Framework

  ✓ You should have automated more than one sprints
  ✓ Ensure you have in-sprint automation concepts introduced as part of project
  ✓ Ensure you have created Regression suite and smoke suite automation
  ✓ Ensure your regression suite consists of minimum 25 test cases and test scripts
  ✓ Ensure your smoke suite covers end to end business functionality

- Trainer-Guided Integrated Project: Robust Test Data Management via TestNG DataProviders
- Trainer-Guided Integrated Project: Comprehensive Execution Control via testng.xml

- Trainer-Guided Integrated Project: Advanced Reporting with Extent Reports & Listeners
- Trainer-Guided Integrated Project: Implementation of Assertions (Hard and Soft)

Day 41

Automation Testing - Selenium with TestNG

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 13	TestNG Concepts	Multi-browser testing using TestNG*
		Rerun failed tests*
		Set Test priority in TestNG
		Dependency in TestNG
		Soft /Hard Assertion in TestNG
		TestNG Report Generation in Selenium WebDriver*
		Customize, PDF & Email TestNG Reports in Selenium WebDriver*
	TestNG Class	TestNG with two test classes
	Data driven	Data driven from XML + POM : Contact Book- POM structure

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Cargo Shipping
- Shopify Registration
- XMLParsing -1
- XMLParsing -2
- XMLParsing -3

Learn the basics of Selenium with Datasource XML Parsing (With XML & JSON)

Rehearse the below Lend a hand enablement given in platform

- XMLParser_Enablement

Followed by refer “Lend-a-Hand” code template with solution

- XML Parsing

Once completing course, Learnings, all Handson & Practice assessments related to Selenium – TestNG skill appear for the Code Challenge Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Code Challenge - Assess-Type-1:Code Challenge - Selenium with TestNG
Assessment Duration: 2 hrs.
Assessment Attempts: 3
No.of Questions appear in each attempt : 1

Project – Mini Project Case Study

- Integrate the mini project.

Day 42 and 43

Automation Testing - Selenium with TestNG

Continuous Learning: Technical Enablement

Learn the basics of Selenium with Datasource JSON Parsing (With XML & JSON)

Rehearse the below Lend a hand enablement given in platform

- JSONParsing_Enablement

Followed by refer “Lend-a-Hand” code template with solution

- JSON Parsing
- Json Solution Explanation

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- JSONParsing – 1
- JSONParsing – 2
- JSONParsing – 3

NOTE: **On Day 42**, Once completing course, Learnings, all Handson & Practice assessments related to Selenium skill appear for the MOCK ICT Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Integrated Capability Test (ICT) - SELENIUM MOCK

Assessment Duration: 2 hrs.
Assessment Attempts: 1
No.of Questions appear in each attempt : 1

Integrated Capability Test (ICT) - SELENIUM

SELENIUM Assess-Type-2: Integrated Capability Test (ICT)

NOTE: **On Day 43**, Once completing course, Learnings, all Handson & Practice assessments related to Selenium skill appear for the Selenium ICT Assessment. This will help you to assess and apply the concepts of the skill learnt in the platform.

Assessment Duration: 4 hrs.
Assessment Attempts: 3
No.of Questions appear in each attempt : 1

Project – Mini Project Case Study

- Integrate the mini project.

Deliver your project and get the review comments from SME.

Day 44 & 45

Evaluation:
Interim Project Evaluation + Technical Evaluation

The overall Interim Technical evaluation will cover topics specified up to the Interim Evaluation in the curriculum and will be conducted via a video interview on the GenC Learn Platform.

There are no Additional Attempts for the Interim Evaluation. RAG Achieving in the first attempt is considered as final RAG for Interim Evaluation.

Day 46

Automation Framework

Click Here to View the BU Mandatory Technical Learning and Handson Document.

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 14	**Test Automation Frameworks**	Introduction to Automation Frameworks
		Data-Driven Testing in Selenium
		Keyword-Driven Testing Framework in Selenium
		Page Factory
		Building a Modular Testing Framework in Selenium
		Hybrid Framework
	Design Patterns (POM)	Understanding Page Object Model (POM) in Selenium
		Creating Page Classes in Selenium
		Implementing Page Factory in Selenium
		Design Patterns in Automation using Selenium (Singleton, Factory etc.)
Chapter 15	**Test Data Handling**	Reading and Writing Data to Files (e.g., Excel, CSV , XML, JSON) using Java
		Test Data Parameterization in Selenium
Chapter 16	**Test Report and Test Log**	Generating Detailed Test Reports (Extent Reports, Allure Reports)
		Screenshot Capture Strategies in Selenium
		Custom Logging with Log4j in Selenium

Continuous Learning: Technical Enablement through Self-learning

After the Trainer-Led Master session on Selenium chapter topics, trainees should:

1. Practice Tekstac Platform Handson and Review respective BU Training Content chapter's PPT and video learning content.
2. Practice DLE (Daily Learning) hands-on exercises provided in the local environment
3. Submit the solutions for evaluation.
4. Refer to and learn the same topics covered by the trainer on Percipio.

This ensures a thorough understanding and application of the Selenium chapter topics.

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 14_Handson
- Chapter 14_PPT Learning Ref.Content
- Chapter 14_Video Learning Ref.Content[Part-1]
- Chapter 14_Video Learning Ref.Content[Part-2]

- Chapter 14_Video Learning Ref.Content[Part-3]

- Chapter 15_Handson
- Chapter 15_PPT Learning Ref.Content
- Chapter 15_Video Learning Ref.Content
- Chapter 16_Handson
- Chapter 16_PPT Learning Ref.Content
- Chapter 16_Video Learning Ref.Content[Part-1]
- Chapter 16_Video Learning Ref.Content[Part-2]
- Chapter 16_Video Learning Ref.Content[Part-3]

Note: You are requested to complete below Integrated Project Deliverable on or before your day 54

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

- Trainer-Guided Integrated Project: Scalable Execution Infrastructure (Selenium Grid & Docker)
- Trainer-Guided Integrated Project: Behavior-Driven Development (BDD) Framework Integration
- Trainer-Guided Integrated Project: Continuous Integration & Delivery (CI/CD) Pipeline Integration

Day 47 to Day 54

Devops

You will be focusing on Devops (Maven, Git, Jenkins, Grid).

Percipio learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Click Here to View the BU Mandatory Technical Learning and Handson Document.

Table of Contents:

Topic #	Topic Name	Topic Objective	References
Behavior-Driven Development (BDD)	Cucumber	Introduction to BDD and Gherkin Syntax	Writing feature files and step definitions.
	Cucumber	Integrating BDD with Selenium Framework	Running scenarios and hooks.
Build Automation & Dependencies	Maven	Introduction to Maven	Understanding Project Object Model (POM).

	Maven	Managing Dependencies and Plugins	Setting up Selenium and other libraries.
	Maven	Executing Tests using Maven Goals	Running tests from the command line.
Version Control System	GIT	Introduction to GIT and GitHub	Basic commands (clone, add, commit, push, pull).
	GIT	Branching and Merging Strategies	Collaborative development practices.
	GitHub	Integrating Test Code with Remote Repository	Pushing the automation framework to the cloud.
Continuous Integration/Delivery	Jenkins	Setting up Jenkins CI Server	Installation and configuration.
	Jenkins	Creating a Jenkins Pipeline	Integrating GIT and Maven projects.
	Jenkins	Scheduling and Triggering Automation Runs	Generating test reports in CI environment.
Distributed Test Execution	Seleniu m GRID	Understanding Selenium Grid Architecture	Hub and Node concepts.
	Seleniu m GRID	Setting up and Configuring the Grid	Running tests in parallel on multiple machines/browsers.
Containerization	Docker	Introduction to Docker	Understanding images and containers.
	Docker	Running Selenium Tests using Docker Containers	Setting up Dockerized Selenium Grid for consistent environments.
	Docker	Creating a Custom Automation Image	Packaging the complete framework.

Day 47, Day 48 & Day 49

Devops (GIT, Maven, Grid, Jenkins)

Continuous Learning: ILT Technical Enablement

Chapter	Chapter Name	Topic
Chapter 17	**Selenium Grid**	Introduction to Selenium Grid
		Setting up Selenium Grid
		Running tests on Selenium Grid
		Achieving Parallel Execution in the Cloud
		Setting up selenium grid in a docker/K8S container
	Distributed Execution	Running Tests on Cloud-Based Selenium Grids (e.g., Sauce Labs, Browser Stack)
		Achieving Parallel Execution in the Cloud
	Maven	Introduction to Maven
		Creating and Managing Maven Projects
		Maven Build Lifecycle (e.g., clean, compile, test, package
	Git Version Control	Introduction to Version Control
		Git Fundamentals (Commits, Branches, Merges)
		Git Workflow (Pull Requests
		Managing Code Repositories on GitHub or GitLab
	Jenkins CI Systems	Introduction to CI/CD and DevOps
		Jenkins Overview and Jobs Configuration
		Integrating Selenium Tests into Jenkins Pipelines
		Groovy Script Introduction and Syntax

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Git Config
- Clone Repo
- Add, Commit and Push
- Pull And Merge
- Merge - Resolve Conflict
- Git Tags
- Build project based on another project
- Build Maven Project (Using Freestyle Project)

Additional Hands-on

- GIT installation & commands implementation
- Install Jenkins and Creation of new job

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 17_Handson
- Chapter 17_PPT Learning Ref.Content
- Chapter 17_Video Learning Ref.Content[Part-1]
- Chapter 17_Video Learning Ref.Content[Part-2]
- Chapter 17_Video Learning Ref.Content[Part-3]
- Chapter 17_Video Learning Ref.Content[Part-4]
- Chapter 17_Video Learning Ref.Content[Part-5]

Day 50 & Day 51

Continuous Learning: Technical Hands-on through Tekstac

Mandatory Hands-on

- Pipeline – compile and test
- Build Web Application
- Compiling Executing Java using POM

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 18_Handson
- Chapter 18_PPT Learning Ref.Content
- Chapter 19_Handson

- Chapter 19_PPT Learning Ref.Content

Additional Hands-on

- Mavenization using Eclipse M2E plugin Options and CLI command

Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 52, Day 53 and Day 54

Continuous Learning: Technical Hands-on

Mandatory Hands-on

- Junit With Maven
- Maven Directory Structure
- Maven Shade Plugin
- Maven App to access External Service
- Capital Service
- Setup Grid and parallel execution

BU Training Content - Chapter Wise [BU Mandatory Technical Learning and Handson]

- Chapter 20_Handson
- Chapter 21_Handson
- Chapter 21_PPT Learning Ref.Content
- Chapter 22_Handson
- Chapter 23_Handson
- Chapter 24_Handson

Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 55 to Day 56

BDD Cucumber

Continuous Learning: Technical Hands-on

Mandatory Hands-on

- Address Book
- Customer Registration

Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 57 to Day 61

Day 57 to Day 61 we will learn API Automation

API Automation

API Testing Fundamentals with Postman

Learnings are recommended in the Platform to understand the fundamental concepts. Apply the concepts learned and solve the Hands-on and Practice Case studies as recommended below.

Note: you are expected to complete below project deliverables on or before day 61

Continuation of Integrated Project Deliverable: Agile and Functional Testing handson through Automation

- Trainer-Guided Integrated Project: The "API Smoke Suite" Postman Collection
- Trainer-Guided Integrated Project: The "Rest Assured" Automation Module (Framework Integration)
- Trainer-Guided Integrated Project: Comprehensive Automation Strategy & Traceability
- Trainer-Guided Integrated Project: Consolidated Execution Reports

Day 57

POSTMAN

Continuous Learning: Technical Hands-on

Mandatory Hands-on

- User Service 1 - Test Using Postman
- User Service 2 - Test Using Postman
- User Service 3 - Test Using Postman
- Cinema Hall Service 1 - Test Using Postman
- Find Winner SOAP Webservice - Test Using Postman

API Automation using RESTASSURED

Continuous Learning: Technical Enablement

REST Assured API Testing: Getting Started with REST Assured Testing

Chapter_1_API_Fundamentals

Mandatory Hands-on

- Verify the Success Status code 200 & 404 for the Bitcoin rate against USD and Non-USD currencies.
- Print Bitcoin rate against USD, GBP & EuR currency

Chapter_2_API_Fundamentals

Mandatory Hands-on

Chapter_2_API using Postman – Part 1

- Download Postman
- Create workspace for API Request
- Create the Collections for API Request
- Sending the first API Request

Chapter_2_API using Postman – Part 2

- Get the response of GET API Request
- Get the response of POST API Request
- Get the response of PUT API Request

Chapter_2_API using Postman – Part 3

- Get the response of PATCH and DELETE API Request
- Create an environment in Postman
- Set the environment variables in Postman

Chapter_2_API using Postman – Part 4

- Get the values of the environment variables in Postman
- Get the ID of the given API in Postman
- Data Parameterization using CSV in Postman
- Test API Using Swagger UI

Chapter_3_API_Fundamentals

Mandatory Hands-on

Chapter_3_API in RestAssured – Part 1

- Verify the response of the GET API

- Validate the 200 status code in Rest Assured
- Validate the 301 status code in Rest Assured
- Validate the 404 status code in Rest Assured

Chapter_3_API in RestAssured – Part 2

- Validate the 500 status code in Rest Assured
- Get the API response in JSON in Rest Assured
- Get the API response in XML in Rest Assured
- Get the API response with single query parameter in Rest Assured

Chapter_3_API in RestAssured – Part 3

- Get the API response with double query parameter in Rest Assured
- Get the API response with path parameter in Rest Assured
- Get the API response with base URI in Rest Assured

Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 58

API Automation using RESTASSURED

Chapter_4_API_Fundamentals

Mandatory Hands-on

Chapter_4_RestAssured – Part 1

- Create workspace for API Request
- Create the Collections for API Request
- Sending the first API Request
- Get the response of GET API Request
- Get the response of POST API Request

Chapter_4_RestAssured – Part 2

- Get the response of GET API Request
- Get the response of PUT API Request
- Get the response of PATCH API Request
- Get the response of DELETE API Request
- Create an environment in Postman

Chapter_4_RestAssured – Part 3

- Set the environment variables in Postman
- Get the values of the environment variables in Postman
- Get the ID of the given API in Postman
- Data Parameterization using CSV in Postman

Chapter_6_API_Fundamentals

Mandatory Hands-on

- Verify the Success Status code 200 for Add product details request
- Print the success response for Add product details request
- Update price of existing product details and verify the success status code 200
- Update price of existing product details and print the updated price from response
- Update only name of the existing product and verify the success code 200 displayed
- Delete the product which is not available in the server and verify the status code 404 and 200

Chapter_7_API_Fundamentals

Mandatory Hands-on

- Verify the response status code with status code method in Rest Assured
- Verify the response status code with assert That in Rest Assured
- Verify the response status code with assertion with status Line method in Rest Assured
- Verify the response status code with assertion with Hamcrest matchers method in Rest Assured
- Handle the response using Getting a single header in Rest Assured
- Handle the response using Getting all headers in Rest Assured
- Validating a header in Rest Assured
- Validating multiple headers in Rest Assured
- Validating JSON schema using matchesJsonSchemaIn Classpath in Rest Assured
- Field-by-Field Validation in Rest Assured
- Structural Validation in Rest Assured
- Validate the inline schema in rest assured
- Validating response using Equality Check in Rest Assured
- Validating response using List Size Check in Rest Assured
- Validating response using Existence Check in Rest Assured

Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 59

API Automation using RESTASSURED

Continuous Learning: Technical Enablement

Chapter_7_API_Fundamentals

Mandatory Hands-on

- Validating response using Multiple Assertions in Rest Assured
- Validating response using Logical Operations in Rest Assured
- Validating response using Custom Parsers in Rest Assured
- Validating response for a List of Values in Rest Assured
- Validate cookies in Rest Assured
- Get the response of an API using single header in Rest Assured
- Get the response of an API using multiple headers in Rest Assured
- Get the response of an API by setting headers from a map in Rest Assured
- Utilize the redirect false command in Rest Assured
- Limit the number of redirects in Rest Assured
- Follow redirects but not use relaxed HTTPS validation on the redirected requests in Rest Assured

Chapter_8_API_Fundamentals

Mandatory Hands-on

- Ensure JSON schema data types are matching for the jsonplaceholder API request.
- Ensure JSON schema data types are not matched for the jsonplaceholder API request.
- Print number of universities in United States

BU Mandatory Handson

- Print number of universities in India
- Print number of universities in Canada
- Print the university names in United States
- Print the university names in India
- Print the university names in Canada
- Store Uganda university details response into a json file
- Store Spain university details response into a text file
- Print second available university name in France
- Verify the United States Population is "318558162" in the year of 2016
- Verify the age is "34" for the name="meelad"

Chapter_9_API_Fundamentals

Mandatory Hands-on

- Verify the success status code and response body from a API using Basic Authentication in Postman
- Verify the success status code and response body from an API using Basic Authentication in RestAssured
- Verify the failure status code and response body from an API using Basic Authentication in Postman

BU Mandatory Handson

- Verify the failure status code and response body from an API using Basic Authentication in RestAssured
- Verify the success status code and response body from an API using Digest Authentication in Postman
- Verify the success status code and response body from an API using Digest Authentication in RestAssured
- Verify the failure status code and response body from an API using Digest Authentication in Postman
- Verify the failure status code and response body from an API using Digest Authentication in RestAssured
- Verify the success status code and response body of an API using API Key Authentication in Postman
- Verify the success status code and response body of an API using API Key Authentication in RestAssured Framework
- Verify the failure status code and response body of an API using API Key Authentication in Postman
- Verify the failure status code and response body of an API using API Key Authentication in RestAssured Framework
- Verify the success status code and response body of an API using Bearer token in Postman
- Verify the success status code and response body of an API using Bearer token in Rest Assured
- Verify the Failure status code and response body of an API using Bearer token in Postman
- Verify the failure status code and response body of an API using Bearer token in Rest Assured
- Verify the success status code and response body of an API using Oauth 2 token in Postman
- Verify if the response is successfully generated for an API using Oauth Authentication type
- Verify if the response is successfully generated for an API using Oauth Authentication type

Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 60

API Automation using RESTASSURED

Chapter_10_API_Fundamentals

Mandatory Handsons:

- Get number of universities in United States and print the response log
- Get number of universities in India and print response log by using ResponseLoggingFilter class
- Get the number of universities in Canada and write response log into text file

BU Mandatory Hands-on

- Get the university names in United States and print response log into ExtentReport
- Verify whether error status code 404 is displayed
- Verify whether except status code 200 is displayed
- Validate the response body for detailed error message. Ensure "404 Not Found" error response message is received
- Log the error details by using ErrorLoggingFilter class
- Handle the Generic exeception and print the same
- Handle the Assertion exeception and print the same.

Chapter_11_API_Fundamentals

Mandatory Hands-on

- Print United States population details for the year 2022.Use Request Specification for Base URI,drilldowns & measures
- Print United States population details for the year 2020.Overwrite Request Specification for Base URI, drilldowns & measures
- Print queried Request Specification by using Query() method

Chapter_12_API_Fundamentals

Mandatory Hands-on

- Print United States population details for the year 2022.Use Response Specification for status code & Content type as JSON
- Print United States population details for the year 2020.Overwrite Response Specification for Status code & ContentType
- Print United States population details for the year 2019. Use ResponseSpecBuilder for Status code & ContentType

Chapter_13_API_Fundamentals

Mandatory Hands-on

- Create a serilization POJO class in Rest Assured framework
- Create a serilization class in Rest Assured framework
- Create a de-serilization POJO class in Rest Assured framework

BU Mandatory Hands-on

- Create a de-serilization class in Rest Assured framework

Chapter_14_API_Fundamentals

Mandatory Hands-on

- Directly hardcode the json body data in the string variable, use it for post request and verify the success status code 200
- Create a separate class file, create a static string variable,store the json data,use it for post request and verify the success status code 200
- Send A JSON file as payload to the request and verify the success status code 200

BU Mandatory Hands-on

- Send A XML file as payload to the request and verify the success status code 200
- Send a request with the Content-Type of form-data parameters and verify the success status code 200
- Upload a png file into server and verify the success status code 200

Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 61

API Automation using RESTASSURED

Chapter_15_API_Fundamentals

Mandatory Hands-on

- Verify the success response of the upload API from postman
- Verify the failure response of the upload API from postman
- Verify the success response of the upload API using multipart from rest assured

BU Mandatory Hands-on

- Verify the success response of the download API from postman
- Verify the failure response of the upload API from postman 1
- Verify the success response of the upload API using multipart from rest assured 1

Chapter_16_API_Fundamentals

Mandatory Hands-on

- Verify the create user API and extract the ID value from Create API in Postman
- Verify if the response values from the API is getting stored in environment variables in postman
- Verify the create user API in RestAssured using faker

BU Mandatory Hands-on

- Verify the create user API in Rest Assured using dynamic data generation such as data from templates
- Verify the create user API in Rest Assured and pass the value of ID using ITest Context
- Verify the Get user API and extract the ID value from Create API in Postman
- Verify the GET user API in Rest Assured and get the value of ID using ITest Context
- Verify the Update user API and extract the ID value from Create API in Postman
- Verify the UPDATE user API in Rest Assured and get the value of ID using ITest Context
- Verify the Delete user API and extract the ID value from Create API in Postman
- Verify the UPDATE user API in Rest Assured and get the value of ID using ITest Context

Chapter_17_API_Fundamentals

Mandatory Hands-on

1. Create a cucumber BDD Framework which should support API and Web automation.

Chapter_18_API_Fundamentals

Mandatory Hands-on

- Working with GitHub and Git
- Integration of Rest assured API test script(Maven/TestNG) Based framework with jenkins Freestyle project
- Execute different methods parallelly using TestNG in Rest Assured

BU Mandatory Hands-on

- Execute different suites parallelly using TestNG in Rest Assured
- Execute different classes parallelly using TestNG in Rest Assured
- Integrate and generate Extent reports in Rest Assured Framework

Project – Hackathon

Integrate the Hackathon project and work on daily basis

Day 62

Docker & Container

Continuous Learning: Technical Enablement

Practice docker through below course
Automated Testing with Docker

GenAI Code Assist (GitHub Copilot)

Comprehensive GenAI Hands-On for Automation Developers

- Trainer-Guided Integrated Project: Chat Endpoint Basics
- Trainer-Guided Integrated Project: Understanding Tokenization & Context Limits(Estimate & Optimize Tokens)
- Trainer-Guided Integrated Project: Prompt Optimization via /ai/chat/with-usage
- Trainer-Guided Integrated Project: Generate Edge Cases
- Trainer-Guided Integrated Project: From User Story to Feature File (AI-Driven BDD Generation)

- Trainer-Guided Integrated Project: Generate Locators for SauceDemo Login (Dynamic Locator Generation)
- Trainer-Guided Integrated Project: AI-Augmented Framework
- Trainer-Guided Integrated Project: Prompt Engineering Game

Project – Hackathon

Deliver your project and get the review comments from SME.

The deliverables of the Hackathon will be evaluated by the BU SME.
Project Evaluation will be based on:

- Source Code
- Functionality Completion, Usage of Features, Code Quality
- Demo of Output

Day 63 - 65

Evaluation:
Final Project Evaluation + Final Technical Evaluation

The overall Final Technical evaluation will cover topics specified in the curriculum and will be conducted through a video interview on the GenC Learn Platform.

Trainees are given two attempts to pass the final evaluation. If a Trainee receives a RAG status of "RED" or "AMBER" status on their first attempt, they have one opportunity to improve and achieve a "GREEN" status on their second attempt.



What is Final Evaluation?

The Final Evaluation will be conducted to certify whether a GenC is eligible to enter into the BU or not. The skill of a GenC will be gauged on the application development and overall technical knowhow towards the end of GenC Training.

Tech SME from BU will be conducting the final tech evaluation. As a fallback, the project mentor can also steer this activity.

The final evaluation will be conducted as two phases. They are the following

1. **Final Technical Evaluation**
2. **Final Project Evaluation**

The mode of these evaluations will be any one of the

lowing:

- F2F(face to face)
- Video Based

1. Final Technical Evaluation (FTE)

The BU Mentor will interview the GenC on various skills achieved throughout the training program and put a score which will be considered for the final PHS of the GenC.

2. Final Project Evaluation (FPE)

In this evaluation, the BU Mentor will be verifying the skills of a GenC on a project perspective. End of this evaluation, the BU Mentor will score the GenC's work based on various evaluation criterions.

How to learn each day?

Each day has a set of learning objectives. These learning objectives can be met by going through the Percipio courses and by completing the hands-on exercises mentioned in the daily plan.

The below strategies will help you decide the learning approach.

Learning Strategy & Approach

Find below few imaginary profiles. For each of these profiles we have defined a recommended learning approach. This is not an exhaustive list. The approaches below might help invent a new way of learning.

Profile #1



Harry Reacher

Engineering Discipline: Electronics

Skills: Python, Ruby on Rails, nginx

Project: Mining Crime Data to get Route Cause Insights

Learning Approach to Programming Languages: I do not want to waste my time learning. I am more practice oriented. I want to work on the problem immediately

What will work for me?

- Directly complete hands on exercises
- Refer Internet or Percipio Courses
- If hands on are implemented early, clarify your friends questions and troubleshoot their issues

Profile #2



Olivia Richards

Engineering Discipline: Computer Science

Skills: Java, C, C++

Project: Library Management System

Learning Approach to Programming Languages: I have interest, but I don't know where to start.

What will work for me?

- Go through the recommended Percipio Course
- Try completing the hands on exercises
- Get your clarifications solved with help from Tech SME
- Get help from other learners in your batch whom had already completed

Profile #3



Greg Anderson

Engineering Discipline: Civil

Skills: C

Project: Fiber reinforced concrete

Learning Approach to Programming Languages: I am scared of programming languages. I haven't got my hands dirty with coding

What will work for me?

- Go through the recommended Percipio Course
- Implement the coding along with the author of the Percipio Course
- Try completing the hands on exercises
- Clarify queries with SME
- Troubleshoot programming issues with help from SME or learner from your classroom whom had already completed

FAQ

1. Who can participate in this program?

   Students who have enrolled for Full Internship Program (or) the Cognizant on-boarded GEN Cs can participate in this program.

2. Is there any pre-learning I should do?

   No. This program is open to all students from any academic discipline.

3. How will I know my RAG status?

   It will be shown to you in the GEN C learn Platform, in your Home Page.

4. Whom do I reach out in case of any queries?

   Coach is your point of contact.

5. What is the significance of Hands-on in the overall learning journey?

   Hands-on focuses on specific topics in a Skill, which you can try and execute in the Platform. Group of such Hands-on exercises will be packaged together as a Code Challenge. This Code Challenge will allow you to benchmark your skills in the learning journey. Hands On/ Code Challenges/ ICT are learning components which will help you in understanding the skills better.

6. What is Code Challenge?
   A problem statement will be provided to you and you need to solve it using a single skill.

7. What is Integrated Capability Test (ICT)?
   A case study problem statement will be provided to you, that you may need solve using the combination of Skills learnt in the given stage.

8. How many attempts are provided for the Coding challenge and ICTs? Is it open all the time for practice?
   The Coding challenges and ICTs are open and there are 3 attempts to take them up.

9. What is the entry criteria for qualifier?
   A minimum of 70% hands-on completion and attempt in the CC & ICT is the eligibility criteria for qualifier.

10. What skills are covered in the qualifier?
    The skills of Stage 1 are covered in the qualifier. Only ONE attempt is provided to clear with a minimum score of 70%

11. What if I fail in the Interim evaluation?
    Your coach will notify your performance in the Interim evaluation. However you can continue with the learning.
12. How many chances will I get in the Final evaluation?
    You'll get 2 chances in the Final evaluation which covers ALL the skills in the learning journey.

13. Whom do I reach out in case of any queries?
    Coach is your point of contact.